

Numerical and Graphical summaries for Categorical Data

Presentation Outline

- 1 Numerical Summaries
- 2 Graphical Summaries
- 3 Important points

Numerical Summaries

- **Frequency Table** (Counts by Treatment Group):

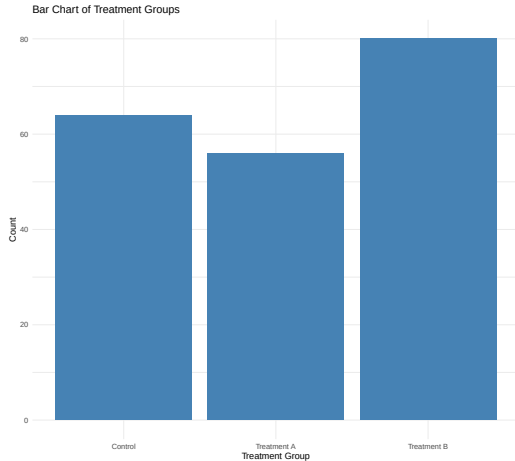
Control	Treatment A	Treatment B
64	56	80

- **Proportions / Relative Frequencies:**

Control	Treatment A	Treatment B
0.32	0.28	0.40

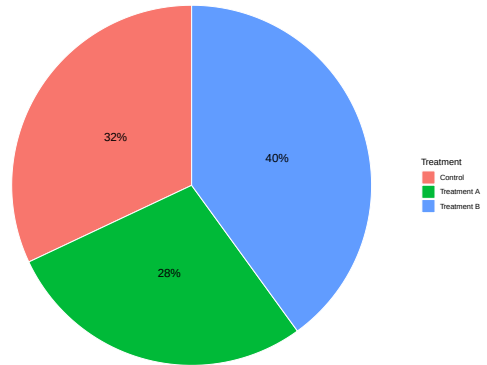
- **Mode:** The most frequently occurring category (here: *Treatment B*)

Bar Charts and Pie Charts



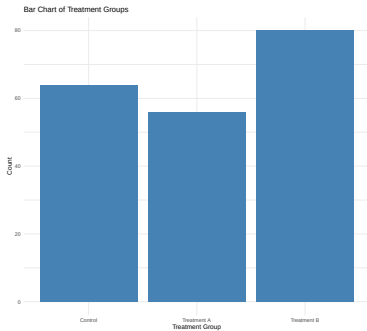
Bar Chart: Treatment Group

Pie Chart of Treatment Group

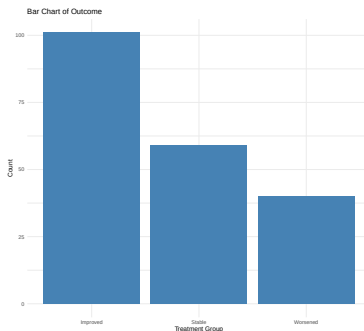


Pie Chart: Treatment Group

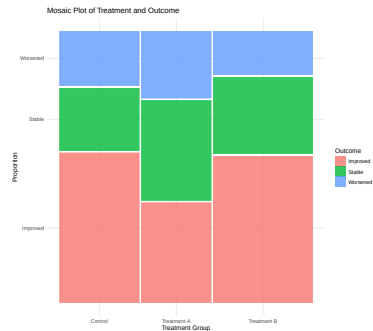
Bar Charts and Mosaic Plot



Treatment Group



Outcome Variable



Mosaic Plot: Treatment $\times$ Outcome

Key Takeaways: Summarizing Categorical Data

- Use frequency tables to count observations in each category.
- Relative frequencies (proportions) are useful for comparing across groups of different sizes.
- The mode is the category with the highest frequency.
- Bar charts and pie charts visualize the distribution of a single categorical variable.
- Mosaic plots visualize relationships between two categorical variables (joint distribution).
- Choose the display method based on whether you're showing one variable or relationships between variables.